Equations

Memo

Section A

QUESTION 1

(a) Solve the following:

(1) $(x-2)(x+1)=0$ (2)

$x-2=0$ or $x+1=0$

$x=2$ or $x=-1$

(3) $(x+2)(x-3)=6$ (4)

$x^2-3x+2x-6-6=0$

$x^2-x-12=0$

$(x-4)(x+3)=0$

$x=4$ or $x=-3$

(b) Solve for x and y simultaneously: (5)

$2x+3y=10 \rightarrow ①$

$x-4y=-6 \rightarrow ②$

Sub into ①

$2(4y-6)+3y=10$

$8y-12+3y=10$

$11y=22$

$y=2$

$x=(4y-6)$

$x=4(2)-6=2$

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Section B

Solve the following equations:

a) $x-3=\frac{18}{x}$ $(x\neq 0)$ (4)

$\frac{x^2-3x}{x}=\frac{18}{x}$

$x^2-3x-18=0$

$(x-6)(x+3)=0$

$x=6$ or $x=-3$

b) $\frac{2x-5}{x+3}-\frac{1}{4-x}=\frac{7}{x^2-x-12}$ (6)

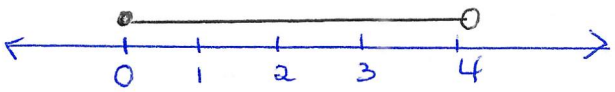
answer on page 3

c) $-2<2-x\leq 2$, represent your answer on a number line. (3)

$-4<-x\leq 0$

$+4>x\geq 0$

$0\leq x<4$



Section C

QUESTION 1

Solve for x:

Answers on page 3

a) (1) $2x^2-7x=-6$ (3)

$2x^2-7x+6=0$

$(2x-3)(x-2)=0$

$2x-3=0$ or $x-2=0$

$2x=3$ $x=2$

$x=3/2$

(2) $\frac{x}{x+2}+\frac{x}{3-x}=\frac{3x-2}{x^2-x-6}$ (4)

(3) $5bx=3ax+d$ (2)

$5bx-3ax=d$

$x(5b-3a)=d$

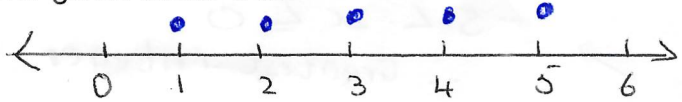
$x=\frac{d}{5b-3a}$

b) (1) $-2<2x-2\leq 8$ (3)

$0<2x\leq 10$

$0<x\leq 5$

(2) Hence, give all the possible solutions to the inequality when it is further given that $x\in\mathbb{Z}$. (2)



QUESTION 2

a) Solve for x and y simultaneously if; (5)

$3x + 2y = 29$ and → ①

$x - 3y + 5 = 0$ → ②

$x = 3y - 5$

Sub 'x' into ①

$3(3y-5) + 2y = 29$

$9y - 15 + 2y = 29$

$11y = 44$

$y = 4$

$\therefore x = 3(4) - 5$

$x = 12 - 5$

$x = 7$

Section D

QUESTION 1

(a) Solve for x:

(3) $x^2 - 3x - 4 = 0$

3) $x^2 - 3x - 4 = 0$

$(x-4)(x+1) = 0$

$x = 4$ or $x = -1$

(4) $(x-2)^2 = 4$

4) $(x-2)(x-2) = 4$

$x^2 - 2x - 2x + 4 - 4 = 0$

$x^2 - 4x = 0$

$x(x-4) = 0$

$x = 0$ or $x = 4$

(2)

(3)

(b) Solve for x and y simultaneously:

$3x - 4y = 5$

$2y - x = -1$

(5)

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Section E

QUESTION 1

(a) Solve for x:

(1) $(x + 2)(x - 3) = 6$

$x^2 - 3x + 2x - 6 - 6 = 0$

$x^2 - x - 12 = 0$

$(x-4)(x+3) = 0 \quad \therefore x = 4$ or $x = -3$

(4)

(3) $t = \frac{y}{5}\sqrt{\frac{p}{x}}$

$5t = y\sqrt{\frac{p}{x}}$

$\frac{5t}{y} = \sqrt{\frac{p}{x}}$

$\left(\frac{5t}{y}\right)^2 = \frac{p}{x}$

$\frac{25t^2}{y^2} = \frac{p}{x}$

$25xt^2 = py^2$

$x = \frac{py^2}{25t^2}$

(3)

(b) Write down the value of the greatest integer that satisfies this inequality

$-6 \leq -2x - 6 < 4$ if $x \in Z$

$0 \leq -2x < 10$

$0 \geq x > -5$

$-5 < x \leq 0$

$\therefore$ Greatest integer is 0

(4)

Section B

b) $\frac{2x-5}{x+3} - \frac{1}{4-x} = \frac{7}{x^2-x-12}$

$\frac{2x-5}{(x+3)} + \frac{1}{(x-4)} = \frac{7}{(x-4)(x+3)}$

Restrictions : $x \neq -3$ & $x \neq 4$

$\frac{(2x-5)(x-4)+(x+3)}{(x+3)(x-4)} = \frac{7}{(x-4)(x+3)}$

$2x^2-8x-5x+20+x+3 = 7$

$2x^2-12x+23-7 = 0$

$2x^2-12x+16 = 0$

$x^2-6x+8 = 0$

$(x-4)(x-2) = 0$

$x \neq 4$ or $x = 2$

See restriction ↑

Section C

Restrictions: $x \neq 3$, $x \neq -2$

a) (2) $\frac{x}{(x+2)} - \frac{x}{(x-3)} = \frac{(3x-2)}{(x-3)(x+2)}$

$\frac{x(x-3)-x(x+2)}{(x+2)(x-3)} = \frac{(3x-2)}{(x-3)(x+2)}$

$x^2-3x-x^2-2x = 3x-2$

$-5x-3x = -2$

$-8x = -2$

$x = \frac{1}{4}$